

Manuscript Supplement V1

Date: 2026-06-05

Target venue: Empirical Software Engineering

Purpose

Provide the submission-facing supplement that supports the main manuscript without forcing the main paper to carry internal run-accounting detail.

Supplement A. Experimental Conditions

Core condition definitions

- **baseline**: the weaker comparison condition that does not receive the reusable protocol artifact being tested
- **taught**: the comparison condition that receives the reusable protocol artifact directly
- **correction**: a two-model condition in which the reviewer critiques and the original executor remains final author
- **takeover**: a two-model condition in which the reviewer or second model becomes the final author
- **quality-bar loop**: a bounded revise-and-check loop with explicit pass and stop conditions

Status labels used in workflow branches

- **pass-core**: the minimum success bar for a workflow artifact chain
- **pass-stretch**: a stronger-than-required success outcome
- **fail-cap**: the loop cap was reached without clearing the required bar
- **blocked**: the run could not proceed cleanly because of a method, runtime, or account-side blocker

Construct note

In this manuscript, **teaching** does not mean durable internal learning. It means that a lesson has been externalized into a reusable protocol artifact and then injected into the taught condition as a structured intervention.

Instruction-specificity note

The current baseline and taught prompts are not length- or specificity-matched. The baseline condition is deliberately weak and underspecified, while the taught condition supplies a reusable protocol artifact and more explicit rubric-aligned output requirements. The present comparison therefore supports an operational claim about explicit protocol scaffolding versus weak prompting. It does not isolate the marginal effect of protocol reuse from the marginal effect of more detailed instructions.

Required follow-on ablation:

- run an instruction-matched control condition that is equally detailed and rubric-aligned but does not invoke the reusable protocol artifact, or invokes an irrelevant protocol artifact

Replication and variance note

Each hosted provider-by-family-by-condition cell in the retained manuscript spine is a single scored run. Clarified reruns and transfer variants are retained as adjacent evidence, not as repeated trials of the same condition. The current evidence base therefore does not support variance estimates, confidence intervals, or significance tests. One- or two-point differences should be interpreted as weak or ceiling-limited unless supported by larger branch-level context.

Supplement B. Branch-Level Run Accounting

Hosted manuscript spine

The hosted manuscript spine is built from OpenAI gpt-4.1, Google gemini-2.5-flash, and Anthropic claude-sonnet-4-6 across:

- BP001 planning
- BP002 structured workflow
- BP003 harder workflow and transfer
- BP004 higher-realism proxy-materialized workflow

The main hosted synthesis uses the current retained comparison sets recorded in:

- docs/benchmarks/cross-provider-synthesis-2026-03-13.md

Open-weight `BP001`

Current branch-level universe:

- five local models
- one baseline and one taught condition per model
- ten scored condition outputs total

Models in the frozen comparison surface:

- qwen2.5:3b
- qwen2.5-coder:3b
- llama3.1:8b
- llama3.2:3b
- deepseek-coder-v2:16b-lite-instruct-q3_K_S

Primary ledger:

- docs/benchmarks/open-weight-run-sheet-v1.md

`BP006` mixture-of-experts branch

Current branch-level universe:

- five eligible models
- twenty ordered executor-reviewer pairs excluding self-review
- six retained and scored pairs
- fourteen planned pairs at the current snapshot

Primary ledger:

- docs/benchmarks/bp006-exhaustive-run-sheet-v1.md

Interpretation rule:

- report BP006 as structured exploratory architecture evidence, not as an exhaustive completed matrix

`BP007B` verifier-backed local planning branch

Current branch-level universe:

- indexed slots T01 through T09
- nine total branch slots

Primary ledger:

- docs/benchmarks/bp007b-run-sheet-v1.md

Interpretation rule:

- methodologically useful diagnostic branch, but not the main local publication surface

`BP008` local workflow branch

Current branch-level universe:

- W01
- W02
- W03

Primary ledger:

- docs/benchmarks/bp008-run-sheet-v1.md

Interpretation rule:

- strongest current local resilience surface

Supplement C. Retained And Superseded Logic

Working definition

Retained means:

- preserved as the current evidence-bearing artifact chain for a run, pair, or branch slot
- not deleted or replaced in the audit trail without documentation
- available for synthesis, review, or manuscript reference

Retained does not mean:

- the only run ever attempted
- the visually strongest result chosen after broad hidden exploration
- the full run universe for a branch

Main retention cases

First stable completed run

Some branches retain the first cleanly completed run as the current evidence-bearing artifact chain.

Clarified rerun replacing an ambiguous earlier run

Some branches required a rerun because an earlier run was not methodologically clean enough for publication use, for example due to:

- packet ambiguity
- clarified placeholder policy
- tightened reviewer logic
- runtime invalidity

In those cases, the earlier run remains in the audit trail and the clarified rerun becomes the current evidence-bearing point.

Exploratory subset

Some branches, especially BP006, use a retained subset for synthesis even though the broader branch is incomplete.

Reporting rule:

- say explicitly that the branch is exploratory or partially complete
- do not let **retained** stand in for exhaustive accounting

Supplement D. Scoring And Review Procedures

Review layers

- L0: initial operator scoring against the benchmark rubric
- L1: operator-side second-pass review or adjudication where applicable
- L2A: model-based secondary review, currently Anthropic on the hosted BP001 and BP002 slice
- L2: distinct human evaluator review, currently completed for the hosted BP001 and BP002 six-run slice

Current review posture

- most of the full portfolio still relies on L0/L1
- operator-side L0/L1 worksheets identify Codex as the evaluator, not a separate human grader
- L2A is used as a secondary consistency check, not as independence proof
- distinct human L2 now exists for the hosted BP001/BP002 slice and is treated as directional support, not condition-level quantitative resolution

Human `L2` handling rule

The Fernanda De La O memo is the controlling human-review artifact. The manuscript uses the memo for directional support and uses the repository-local score-table transcription only for column totals. The score table records one column per provider-by-benchmark cell rather than separate baseline and taught scores, and its absolute totals are lower than the operator-side taught scores. The paper therefore does not use the memo as condition-level quantitative confirmation and does not rely on unlabeled rubric rows for finer-grained numerical interpretation.

Model-based `L2A` handling rule

The Anthropic L2A review is a useful secondary consistency check, but it is not treated as independent adjudication. It uses `claude-sonnet-4-6` as judge, including for an Anthropic-authored output in the hosted slice, so same-family and partial self-judging risks remain.

Supplement E. Results Tables

Hosted cross-provider within-model uplift

Provider	Model	BP001 baseline	BP001 taught	BP002 baseline	BP002 taught	BP003 baseline	BP003 taught
---	---	---	---	---	---	---	---
Open AI	<code>gpt-4.1</code>	3/10	9/10	5/10	9/10	8/10	10/10
Google	<code>gemini-2.5-flash</code>	3/10	7/10	8/10	9/10	8/10	10/10
Anthropic	<code>claude-sonnet-4-6</code>	1/10	8/10	8/10	9/10	8/10	10/10

Local evidence ladder

Surface	Strongest retained result	Interpretation
---	---	---
Open-weight BP001	taught ceiling 6/10	structural planning uplift exists, but absolute planning fidelity remains limited
BP007B verifier-backed planning	T07 reaches pass-core without score uplift	stronger verification removes false positives but does not rescue weak local planning
BP008 workflow branch	W01 and W03 reach pass-core	concrete workflow tasks preserve the clearest local resilience effect

`BP006` architecture summary

Pair	Executor	Reviewer	Best condition
---	---	---	---
P1	<code>gpt-4.1</code>	<code>claude-sonnet-4-6</code>	correction
P2	<code>gemini-2.5-flash</code>	<code>gpt-4.1</code>	takeover
P3	<code>gpt-4.1</code>	<code>gemini-2.5-flash</code>	no material MoE gain
P4	<code>claude-sonnet-4-6</code>	<code>gpt-4.1</code>	correction
P5	<code>deepseek-coder-v2:16b-lite-instruct-q3_K_S</code>	<code>gpt-4.1</code>	takeover
P6	<code>deepseek-coder-v2:16b-lite-instruct-q3_K_S</code>	<code>llama3.2:3b</code>	modest local-local gain

Supplement F. Local Runtime And Environment Constraints

The local branch should be interpreted with the following environmental constraints in mind:

- one-model-at-a-time storage discipline on the current machine
- slower local inference than hosted runs
- lower-quality all-local review loops than hosted-review loops
- stronger sensitivity to runtime instability during revise-and-check cycles

These constraints matter because local ceilings reflect both model capability and operating environment. The local branch therefore supports resilience claims, not hosted-quality parity claims.

Supplement G. Known Evidence Gaps And Follow-On Work

Current evidence gaps

- The protocol-versus-instruction-specificity effect is not isolated because no instruction-matched non-protocol control has been run.
- Hosted cells are single retained runs, so no run-to-run variance or statistical significance estimate is available.
- The Anthropic BP001 baseline is retained as an operator narrative rather than a full verbatim transcript.
- Human L2 review exists for the hosted BP001/BP002 slice, but it does not provide separate baseline-versus-taught score columns.

Required major-revision work before stronger claims

1. Add an instruction-matched control condition.
2. Repeat each retained condition enough times to report variance and avoid treating one- or two-point deltas as stable effects.
3. Preserve verbatim transcripts for every future baseline and taught output.
4. Expand distinct human review or redesign the human review worksheet to score baseline and taught conditions separately.

Supplement H. Data, Materials, And Audit Trail Availability

The following materials are publicly available as part of the manuscript evidence chain in the Protocol-Scaffolded Agentic Workflows Artifacts repository:

- repository: <https://github.com/Quantyra/protocol-scaffolded-agentic-workflows>
- fixed paper-support release: <https://github.com/Quantyra/protocol-scaffolded-agentic-workflows/releases/tag/v1.0.7>
- benchmark packets
- run sheets
- retained artifact chains
- synthesis memos
- scoring rubrics
- human and model-based hosted review materials

The separate EMSE title-page declarations now point directly to the same public repository and fixed release.